

MIE230: Engineering Analysis

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1 Multivariable Functions

A uni-variable function is where a function depends on a single independent variable. Essentially, $y = f(x)$ for $f : \mathbb{R} \rightarrow \mathbb{R}$. Consider a projectile with initial velocity v at angle θ . Then:

$$\Delta x = v^2 \sin(2\theta)/g$$

This is a multivariable function! $\Delta x = f(v, \theta)$ where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Generally, we express multivariable functions as $y = f(x_1, x_2, \dots)$. For 2D, we express this as $z = f(x, y)$.

Lines of equal z value are called *iso-lines* or *level curves*. Consider a paraboloid: $z = x^2 + y^2$, this is a 3D surface. This looks similar to equation of a circle in $\mathbb{R}^2$. In fact, $z = r^2$. For a given value of z , its level curve is a circle.

Now, consider a 3D scalar field, $T(x, y, z)$. Suppose we want to know where in our space is the temperature some constant $T = c$. Then, $T(x, y, z) = c$ yields a surface! This is called an iso-surface which is analogous to iso-lines. For example, if

$$T(x, y, z) = z - x^2 - y^2 \implies z = x^2 + y^2 + c$$

here z is an iso-surface of T . Essentially, iso-lines are created through cutting the surface with a lower dimensional plane and tracing the function on that plane.

Now what about lines of constant x and y ?

Consider the distance function $\Delta x = v^2 \sin(2\theta)/g$. If we are solely interested in the velocity, we let θ be a constant. Hence, the equation takes the form:

$$\Delta x = v^2 \cdot \alpha$$

with α being some constant. This is a line. It is called a *trace* of Δx . We can also do a similar analysis for θ :

$$\Delta x = \sin(2\theta) \cdot \alpha$$

These are essentially the intersections of different planes with our multivariable function. This idea is fundamental to understanding partial derivatives.

1.1 Example

Suppose you invest money in an account that pays 5% interest compounded continuously. If you invest P dollars in the account, the amount of A of money in the account after t years is given by

$$A = Pe^{0.05t}$$

The variables P and t are independent of each other, so using functional notation we write

$$A(P, t) = Pe^{0.05t}$$

1. Find the amount of money in the account after 7 years if you originally invest 1000 dollars.
2. Evaluate $A(5000, 8)$. Explain in words what this calculation represents.
3. Now consider only the situation where the amount invested is fixed at 1000 dollars. Calculate the amount of money in the account after t years.
4. Now consider the situation where we want to know the amount of money in the account after 10 years given various initial investments. Calculate the amount of money in the account.

2 Partial Derivatives

Consider a univariable function $f : \mathbb{R} \rightarrow \mathbb{R}$. We can approximate its derivative using:

$$\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Geometrically, this is the slope of the secant line between the two points of distance Δx . Then, the interpretation of the derivative is the slope of the tangent line to the function:

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

This is the actual slope of f , denoted $f'(x)$. Now, let us consider a function of several variables $z = x^2 \sin(y)$. Consider the following limits:

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta z}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{z(x + \Delta x, y) - z(x, y)}{\Delta x} = \frac{\partial z}{\partial x}$$

$$\lim_{\Delta y \rightarrow 0} \frac{\Delta z}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{z(x, y + \Delta y) - z(x, y)}{\Delta y} = \frac{\partial z}{\partial y}$$

3 Chain Rule and Parametrization

Sometimes, it's easier to do a problem in a different coordinate system than the one provided by the problem. This demands the use of parametrization. We will then use this in the chain rule to do calculus with our parametrized coordinates.

Consider the function $y(t) = \sin(t^2)$. To differentiate this, we essentially break it up and use the chain rule such that:

$$y(x) = \sin(x), \quad x(t) = t^2, \quad \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = 2t \cos(t^2)$$

We apply a similar principle to multivariable functions. What we are really doing with the chain rule is using differentials:

$$dy = \frac{dy}{dx} dx, \quad dx = \frac{dx}{dt} dt \implies dy = \frac{dy}{dx} \frac{dx}{dt} dt$$

This is where the chain rule comes from. Consider a multivariable function $z = f(x, y)$. Then, the chain rule can be written as:

$$\boxed{df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy}$$

Now if we apply the above parametrization, we find that $z = f(x, y) = f(x(t), y(t)) = f(t)$. Thus,

$$dz = \left(\frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \right) dt \quad dz = \frac{dz}{dt} dt \implies \frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

This is the multivariable chain rule.

Example Suppose that the temperature of a metal plate is given by the function

$$T(x, y) = 100 - (x^2 + 4y^2)$$

where temperature is measured in degrees Fahrenheit and x and y are each measured in feet.

1. Find T_x and T_y and their units.

Ans:

$$\frac{\partial T}{\partial x} = -2x^\circ F/ft \quad \frac{\partial T}{\partial y} = -8y^\circ F/ft$$

2. Suppose an ant is walking along the x -axis at a rate of 2 feet per minute towards the origin. When the ant is at the point $(2, 0)$, what is the instantaneous rate of change in the temperature dT/dt that the ant experiences?

Ans:

$$\begin{aligned} \frac{dx}{dt} &= -2 \text{ ft/min}, \quad \frac{dy}{dt} = 0 \text{ ft/min} \\ \implies \frac{dT}{dt} &= \frac{\partial T}{\partial x} \frac{dx}{dt} = 8 \text{ ft/min} \end{aligned}$$

3. Suppose instead the ant walk along an ellipse with $x = 6 \cos(t)$ and $y = \sin(3t)$, where t is in minutes. Find dT/dt at $t = \pi/6, \pi/4$ and $\pi/3$. What does this seem to tell you about the path along which the ant is walking?

Ans:

$$\frac{dT}{dt} = \frac{\partial T}{\partial x} \frac{dx}{dt} + \frac{\partial T}{\partial y} \frac{dy}{dt}$$

At these times, the temperature is all 0. Use matlab to see.

Sometimes it is easier to work in polar coordinates! Suppose $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$.

Then:

$$\begin{pmatrix} \frac{\partial f}{\partial r} \\ \frac{\partial f}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} \quad \text{a.k.a} \quad \boxed{z^m = J(r, \theta) z^n}$$

Now suppose we parametrize a surface with 2 independent variables (u, v) . Then there's no full derivative, but partial derivatives with respect to u and v .

4 Double Integrals

Recall, a single integral calculates the area under the curve of some function $f : \mathbb{R} \rightarrow \mathbb{R}$. This area is signed and can be approximated through Riemann sums. Recall:

$$R_n = \sum_{i=1}^n f(x_i) \Delta x$$

Where Δx is the width of a rectangle and $f(x_i)$ is the height of each rectangle. Will learn more smart ways of approximating integrals in MIE235. In the limit as $n \rightarrow \infty$, the Riemann sum becomes an integral.

Double integrals find the volume under a surface $z = f(x, y)$ and can be estimated with a double Riemann sum:

$$\sum_{j=1}^n \sum_{i=1}^m f(x_{ij}, y_{ij}) \Delta x \Delta y$$

As $(m, n) \rightarrow (\infty, \infty)$, we find the double integral:

$$\iint_R f(x, y) dA \quad \text{or} \quad \iint_R f(x, y) dx dy$$

Although in different coordinates, $dA \neq dx dy$. R is the region on the xy -plane. Remember that this is a signed volume. Properties:

$$\begin{aligned} \iint_R k f(x, y) dA &= k \iint_R f(x, y) dA \\ \iint_R f + g dA &= \iint_R f dA + \iint_R g dA \\ \iint_R f(x, y) dA &= \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA \end{aligned}$$

The piecewise integration property has extreme application in finite element methods.

In 2D, 3D and 4D, the order of integration does not matter:

$$\boxed{\iint_R f(x, y) dx dy = \iint_R f(x, y) dy dx}$$

This comes from the property of Riemann sums.

Example let $f(x, y) = x + y^2$ on a rectangular region $[0, 2] \times [0, 3]$. Calculate $\iint_R f(x, y) dA$.

Ans:

$$\iint_R f(x, y) dA = \int_0^2 \int_0^3 x + y^2 dy dx = \int_0^2 3x + 9 dx = 6 + 18 = 24$$

What if we reverse the order of integration ?

$$\iint_R f(x, y) dA = \int_0^3 \int_0^2 x + y^2 dx dy = \int_0^3 2 + 2y^2 dy = 6 + 18 = 24$$

So the order of integration does not matter. In some cases, it can be useful to switch the order of integration.

Suppose we want to find the double integral $\iint_D x^2 y dA$ over a non-rectangular region, but a triangular one as shown in slide 8.

At each y value, we have a different x value. The y values can actually be written as a function of x . $y(x) = \frac{3}{2}x$. So the region is between $y = 0$ and $y = \frac{3}{2}x$, and $x \in [0, 2]$. So:

$$\iint_D x^2 y dA = \int_0^2 \int_{y=0}^{y=\frac{3}{2}x} x^2 y dy dx = \int_0^2 x^2 \int_{y=0}^{y=\frac{3}{2}x} y dy dx = \int_0^2 \frac{9}{8} x^4 dx = \frac{9}{8} \times \frac{32}{5} = \frac{36}{5}$$

What if we flip the integral ? Then $y \in [0, 3]$ but $x \in [\frac{2}{3}y, 2]$.

$$\iint_D x^2 y dA = \int_0^3 y \int_{x=\frac{2}{3}y}^2 x^2 dx dy = \int_0^3 \frac{8}{3} y - \frac{8}{81} y^4 dy = \frac{8 \cdot 9}{3 \cdot 2} - \frac{8 \cdot 243}{81 \cdot 5} = \frac{36}{5}$$

If the region of integration is bound by two general curves $y = h(x)$ and $y = g(x)$ and $x \in [a, b]$, then:

$$\iint_R f(x, y) dA = \int_a^b \int_{y=g(x)}^{y=h(x)} f(x, y) dy dx$$

Similarly, let $p(y), q(y)$ be the inverses of $h(x), g(x)$ respectively. And let $y \in [c, d]$ Then:

$$\iint_R f(x, y) dA = \int_c^d \int_{x=p(y)}^{x=q(y)} f(x, y) dx dy$$

Consider finding the area of a semi circle of radius 3 with a double integral by assuming $f(x, y) = 1$:

$$\int_{-3}^3 \int_{x=0}^{x=\sqrt{9-y^2}} dx dy = \int_{-3}^3 \sqrt{9-y^2} dy = \left[\frac{9 \arcsin(y/3) + y\sqrt{9-y^2}}{2} \right]_{y=-3}^{y=3} = \frac{9\pi}{2}$$

Now, it really doesn't make sense to integrate a semi-circle in cartesian. SO how can we do this in polar coordinates where $x = r \cos \theta$ and $y = r \sin \theta$?

$$f(x, y) = f(r \cos \theta, r \sin \theta) = f(r, \theta)$$

That's the easy part. Now, how can we turn our dA into $dr d\theta$ form ? The polar rectangle, is actually dependent on r . This comes from the Jacobian (next lecture). So:

$$\iint_R f(x, y) dA = \iint_R f(r, \theta) r dr d\theta$$

So if we find the area of our semi-circle with polar coordinates:

$$A = \int_{-\pi/2}^{\pi/2} \int_{r=0}^{r=3} r dr d\theta = \int_{-\pi/2}^{\pi/2} \frac{9}{2} d\theta = \frac{9\pi}{2}$$

Note, if $f(x, y) = p(x)q(y)$ then:

$$\iint f(x, y) dx dy = \int p(x) dx \int q(y) dy$$

5 Applications of Double Integrals

5.1 Fundamental Theorem of Double Integrals, Indefinite Integrals

Analogous to the fundamental theorem of calculus (FTOC) in 1 dimension, we also have for 2 dimensions:

$$\frac{\partial}{\partial x} \int f(x, y) dx = f(x, y)$$

$$\frac{\partial}{\partial y} \int f(x, y) dy = f(x, y)$$

$$\frac{\partial^2}{\partial x \partial y} \iint f(x, y) dx dy$$

We can also take indefinite double integrals analogous to indefinite single integrals.

5.2 Applications

5.2.1 Average Value

Suppose we have a function of several variables $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then the average value of this function over some domain R , is given by:

$$\bar{f}_{2d} = \frac{\iint_R f(x, y) dA}{\iint_R dA} = \frac{1}{L} \iint_R f(x, y) dA = \frac{V}{A}$$

Recall that for a single variable, the average value is given by:

$$\bar{f}_{1d} = \frac{\int_a^b f(x) dx}{\int_a^b dx} = \frac{1}{b-a} \int_a^b f(x) dx = \frac{A}{L}$$

Example Suppose the temperature over a region is given by $T(x, y) = 100 - (x^2 - 4y^2)$. R is bounded by $(x, y) \in [-8, 8] \times [-4, 4]$. What is the average value ?

$$\bar{T} = \frac{\iint_R T dA}{\iint_R dA}$$

Since the function is symmetric, we can change the numerator to:

$$\iint_R T \, dA = 4 \int_0^4 \int_0^8 100 - (x^2 - 4y^2) \, dx \, dy \approx 7339$$

$$\iint_R dA = 16 \cdot 8 = 128 \implies \bar{T} = \frac{7339}{128} \approx 57.4^\circ F$$

5.2.2 Density and Mass

Suppose we have a region of 2 dimensions, R with a variable density $\rho(x, y)$, the mass of the region is then given by:

$$m = \iint_R \rho(x, y) \, dA$$

for a uniform density, we would have $m = \rho \iint_R dA$. If we want the average density,:

$$\bar{\rho} = \frac{\iint_R \rho(x, y) \, dA}{\iint_R dA} = \frac{1}{A} \iint_R \rho(x, y) \, dA = \frac{m}{A}$$

5.2.3 Center of Mass and Moments of Inertia

Suppose we have a set of discrete points:

$$S = \bigcup_{n=1}^4 (x_n, y_n)$$

If these points are connected, and we apply a moment or a force to this, $M = F \times r$. For point 1:

$$M_1 = m_1 g(x_1 - \bar{x}) \implies \sum M = \sum_{i=1}^4 m_i g(x_i - \bar{x}) = 0$$

The center of gravity is given by:

$$\frac{\sum m_i g x_i}{\sum m_i g} = \bar{x} \quad \frac{\sum m_i g y_i}{\sum m_i g} = \bar{y}$$

For a continuous domain, the summations turn into integrals. The continuum version for center of mass becomes:

$$\bar{x} = \frac{\iint_R x \rho(x, y) dA}{\iint_R \rho(x, y) dA} = \frac{1}{M} \iint_R x \rho(x, y) dA \quad \bar{y} = \frac{\iint_R y \rho(x, y) dA}{\iint_R \rho(x, y) dA} = \frac{1}{M} \iint_R y \rho(x, y) dA$$

Suppose we're not worried about varying gravitational fields and we have uniform density:

$$\bar{x} = \frac{\iint_R x dA}{\iint_R dA} = \frac{1}{A} \iint_R x dA \quad \bar{y} = \frac{\iint_R y dA}{\iint_R dA} = \frac{1}{A} \iint_R y dA$$

The integrals are called the first moments of area. This is because x and y are raised to the power of 1. The second moment would turn x and y to x^2 and y^2 .

Example Calculate mass and centroids of a quarter unit circle, assuming $\rho = 1$.

$$\begin{aligned} M &= \iint_R \rho dA = \iint_R dA = \frac{\pi}{4} \\ \implies \bar{x} &= \frac{1}{M} \iint_R x dA = \frac{4}{\pi} \int_0^{\pi/2} \int_0^1 r^2 \cos \theta dr d\theta = \frac{4}{3\pi} \\ \bar{y} &= \frac{1}{M} \iint_R y dA = \frac{4}{\pi} \int_0^{\pi/2} \int_0^1 r^2 \sin \theta dr d\theta = \frac{4}{3\pi} \end{aligned}$$

Moments of inertia about x and y axes:

$$I_x = \iint_R y^2 \rho(x, y) dA \quad I_y = \iint_R x^2 \rho(x, y) dA$$

These are called the second moments of area since x, y are raised to the power of 2. The polar moment of inertia (about the z axis) is:

$$I_0 = \rho \iint_R r^2 dA = \rho \iint_R x^2 + y^2 dA = I_x + I_y$$

This is assuming constant density. Otherwise, the density function will also be a part of the integrand.

5.3 Change of Variables in Double Integrals

How do we integrate in arbitrary coordinates ? How do we integrate in a coordinate system that may be more convenient ? Recall:

$$\int f(x) dx = \int f(x(t)) \frac{dx}{dt} dt$$

For double integral parametrized from (x, y) to (u, v) :

$$\begin{aligned} dx &= \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv \\ dy &= \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv \\ \Rightarrow \begin{Bmatrix} dx \\ dy \end{Bmatrix} &= \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \begin{Bmatrix} du \\ dv \end{Bmatrix} = [J] \begin{Bmatrix} du \\ dv \end{Bmatrix} \\ dx dy &= |J_{det}| du dv = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv \end{aligned}$$

Note, here $|J|$ denotes the absolute value of $\det(J)$. J is called the Jacobian matrix and $|J|$ is the absolute value of its determinant. To generalize change of coordinates in double integrals, we use the Jacobian:

$$\iint_R f(x, y) dA = \iint_{R^*} f(u, v) |\mathcal{J}(u, v)| du dv$$

So now we can see where the transformation of integrals to polar coordinates comes from:

$$\mathcal{J}(r, \theta) = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix} \Rightarrow |J| = r$$

Example Evaluate the following double integral over the diamond region with vertices at $(0, 0), (1, 1), (-1, 1), (2, 0)$:

$$\iint_R x^2 + y^2 dx dy$$

This would be very, very messy in cartesian coordinates. The region is bounded by the lines $y = x, y = -x, y = -x + 2, y = x - 2$. Let's define a new parametric space $u = x + y, v = x - y$:

$$x = \frac{1}{2}(u + v) \quad y = \frac{1}{2}(u - v) \implies |J| = \frac{1}{2}, \quad f(x, y) = x^2 + y^2 = \frac{1}{2}(u^2 + v^2)$$

Now the limits of the integral become:

$$\int_{u=0}^{u=2} \int_{v=0}^{v=2} \frac{1}{2}(u^2 + v^2) \cdot \left(\frac{1}{2}\right) dv du = \frac{8}{3}$$

The Jacobian has great significance in Finite Element Methods by transforming mesh elements into better coordinate system using the Jacobian Diagnostics software to evaluate integral equations easier.

6 Triple Integrals

Similar to double integrals, the mass contained in a volume with density $\rho(x, y, z)$ can be estimated using a triple Riemann sum:

$$\sum_{k=1}^n \sum_{j=1}^m \sum_{i=1}^l \rho(x_{ijk}, y_{ijk}, z_{ijk}) \Delta x \Delta y \Delta z$$

For volume of a solid S :

$$V = \iiint_S dV$$

The average value of a function is also:

$$\bar{f} = \frac{\iiint_S f(x, y, z) dV}{\iiint_S dV} = \frac{1}{V} \iiint_S f(x, y, z) dV$$

The same principles also apply to moments of inertia and centers of mass/gravity.

Example Evaluate the triple integral of $f(x, y, z) = x - y + 2z$ over the box $B = [-2, 3] \times [1, 4] \times [0, 2]$.

$$\iiint_B f(x, y, z) dV = \int_0^2 \int_1^4 \int_{-2}^3 x - y + 2z dx dy dz = 0$$

To visualize a scalar field, we can use *iso-surfaces*! Let $f(x, y, z) = \text{const}$ then graph the resulting function or field! Why is this integral 0? Because when we graph the iso-surfaces, we find that the iso-surfaces are planes that are increasing, so the planes cancel each other out.

Given a tetrahedron in the first octant, bounded by the coordinate planes and the face plane $x + 2y + 3z = 6$, calculate the mass if $\rho(x, y, z) = x + y + z$.

$$M = \int_{x=0}^{x=?} \int_{y=0}^{y=?} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \rho(x, y, z) dz dy dx$$

Let

$$\int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \rho(x, y, z) dz = F(x, y)$$

then

$$\int_{x=0}^{x=6} \int_{y=0}^{y=3-\frac{1}{2}x} F(x, y) dy dx = \int_{x=0}^{x=6} \int_{y=0}^{y=3-\frac{1}{2}x} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \rho(x, y, z) dz dy dx$$

Let S be the solid cone bounded by $z = \sqrt{x^2 + y^2}$, $z = 3$. Calculate the mass of the solid cone.

If we think about the z bound, it is just circles with radius z since $z^2 = x^2 + y^2$. So we may be better off using cylindrical coordinates.

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

then $|J(r, \theta, z)| = r$:

$$\iiint_S f(x, y, z) dV = \iiint_{S^*} f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

Note that r may be dependent on z !. Assume $\rho(x, y, z) = 1 + z$:

$$M = \int_{z=0}^{z=3} \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=z} (1 + z) r dr d\theta dz = \frac{351\pi}{12} \approx 91.9$$

To do a sanity check, we can use the volume of a cone and the average value of the density to see if our answer makes sense! $\bar{\rho} \approx 3.25$ and $V = \frac{\pi R^3 h}{3} = 9\pi \approx 28.3$. Since $M = \bar{\rho}V \approx 91.97$ which means our answer is very likely to be right.

There is another coordinate system we can use to perform integration which is spherical coordinates:

$$x = r \sin \phi \cos \theta \quad y = r \sin \phi \sin \theta \quad z = r \cos \phi$$

$$r = \sqrt{x^2 + y^2 + z^2} \quad \phi = \cos^{-1}(z/r) \quad \theta = \tan^{-1}(y/x)$$

The Jacobian for this integral is $|J(r, \theta, \phi)| = r^2 \sin \phi$. In motion problems sometimes it is much more convenient to use spherical coordinates than cartesian.

7 Review of Vectors

Vectors have both magnitude and direction. It is visualized through an arrow or a directed line segment. Examples are force, velocity, momentum, etc. Length of vector indicates magnitude and orientation and arrowhead indicate direction. Start is called the tail and the end is called the head.

Vectors having the same magnitude and direction are considered identical regardless of location!

In 2D, a vector is defined by two sets of (x, y) coordinates. If:

$$Q = (1, 1), P = (4, 5)$$

Then QP is the vector from Q to P. A vector starting at the origin is called a position vector. Head minus tail gives vector components:

$$P - Q = (4, 5) - (1, 1) = (3, 4)$$

In matlab, the input $[x_1, x_2, x_3, \dots, x_n]$ is called a vector or 1D array.

Too boring to write rest of vectors, basically dot product and cross product, refer to the "Geometry of Vectors, Lines and Planes" handout. The volume of a tetrahedron formed by three edge vectors $\vec{a}, \vec{b}, \vec{c}$, is given by:

$$V = \frac{1}{6}[a \cdot (b \times c)]$$

Since volume of a tetrahedron is 1/6 the volume of a parallelepiped.

The moment of a force is given by:

$$\vec{m} = \mathbf{r} \times \vec{p}$$

8 Vector-Valued Functions

Recall, a scalar function outputs a single real value given some number of real inputs. Essentially, $f = f(P)$ where P is a point in our manifold.

Vector functions output a vector value given some number of real input parameters.

$$\vec{v}(P) = \begin{bmatrix} v_1(P) \\ v_2(P) \\ v_3(P) \end{bmatrix}$$

again, with P being a point in our physical space. v_1, v_2, v_3 are yet to be assigned to a coordinate system since vector magnitude and direction is not dependent on our choice of coordinates! We will talk about 3 types of vector functions; parametric curve in space $\mathbf{r}(t)$, surface normals $\vec{n}(u, v)$, and vector fields $\vec{F}(\mathbf{r})$. For now,

$$\vec{v}(P) = \vec{v}(x, y, z) = v_1(x, y, z)\hat{i} + v_2(x, y, z)\hat{j} + v_3(x, y, z)\hat{k}$$

Consider the velocity field on a rotating disk:

$$\vec{v} = \vec{\omega} \times \mathbf{r} = (\omega\hat{k}) \times (x\hat{i} + y\hat{j} + z\hat{k}) = -\omega y\hat{i} + \omega x\hat{j}$$

$$\implies v_1(x, y) = -\omega y, \quad v_2(x, y) = \omega x$$

Consider a parametric curve:

$$\mathbf{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

The derivative of $\mathbf{r}$ is defined as:

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

For example, suppose $\vec{v}(t) = [t, t^2, 0]$. Then $\vec{v}' = [1, 2t, 0]$. Vector derivatives are closed under addition. Here are some properties:

$$[\vec{u}(t) + \vec{v}(t)]' = \vec{u}'(t) + \vec{v}'(t)$$

$$[\vec{v}(f(t))]' = \vec{v}'(f(t)) \cdot f'(t)$$

$$[c\vec{v}(t)]' = c \cdot \vec{v}'(t)$$

$$[f(t)\vec{v}(t)]' = f'(t)\vec{v}(t) + f(t)\vec{v}'(t)$$

$$[\vec{u} \cdot \vec{v}]' = \vec{u}' \cdot \vec{v} + \vec{u} \cdot \vec{v}'$$

$$[\vec{u} \times \vec{v}]' = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$$

What if we have $\vec{n}(u, v)$? Then we have partial derivatives:

$$\frac{\partial \vec{n}}{\partial u} = \partial_u x_j e^j$$

8.1 Parametric Curves

Parametric curves are more general than projecting functions onto planes. Recall the equation of a line:

$$\mathbf{r}(t) = \vec{a} + \vec{b}t$$

$\vec{a}$ is the position vector at the start of the line, $\vec{b}$ is the vector parallel to the line or along the line and t is the scalar parameter of location along the line. We can also write this as:

$$\mathbf{r}(t) = (a_1 + b_1 t)\hat{i} + (a_2 + b_2 t)\hat{j} + (a_3 + b_3 t)\hat{k}$$

Consider the equation of an ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This is very awkward to plot. But we can also express this in parametric form:

$$\mathbf{r}(t) = [a \cos t, b \sin t]$$

this is much easier to work with and interpret. What about twisted curves ? An example is the circular helix:

$$\mathbf{r}(t) = a \cos t \hat{i} + a \sin t \hat{j} + ct \hat{k}$$

a is the radius of the helix and c is the pitch. A parametric curve is simply tracing the head of a position vector. A simple curve is that which does not intersect itself.

9 Differential Geometry of Curves

9.1 Tangent Vector

Consider a parametric curve $\mathbf{r}(t)$. Then the tangent vector to this curve is defined as:

$$\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

This line just touches the curve at a given point. The unit tangent vector is defined as:

$$\vec{u}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Note, active calculus textbook calls this vector $\vec{T}'(t)$. Consider the elliptic curve:

$$\mathbf{r}(t) = 2 \cos t \hat{i} + \sin t \hat{j}$$

then the tangent vector is:

$$\mathbf{r}'(t) = -2 \sin t \hat{i} + \cos t \hat{j}$$

9.2 Arc Lengths

Break a curve $\mathbf{r}(t)$ up into discrete pieces. Then the i th arc length is :

$$l_i = |\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})|$$

So the total length is a sum over this, and for higher accuracy, we take the limit as $n \rightarrow \infty$:

$$l = \lim_{n \rightarrow \infty} \sum_{i=1}^n |\mathbf{r}(t_i) - \mathbf{r}(t_{i-1})|$$

If we multiply top and bottom by Δt , we can define the term in the absolute value to a derivative:

$$l = \lim_{n \rightarrow \infty} \sum_{i=1}^n |\mathbf{r}'(t)| \Delta t = \int_a^b \sqrt{\mathbf{r}' \cdot \mathbf{r}'} dt$$

Consider a general ellipse:

$$\mathbf{r}(t) = [a \cos t, b \sin t] \implies \mathbf{r}'(t) = [-a \sin t, b \cos t]$$

The total perimeter for this ellipse results in an elliptic integral! There is no closed form solution, so Ramanujan used Taylor expansions to approximate. The question of interest is the arc length function; depending on t how far along the curve am I?

$$s(t) = \int_{\tau=0}^{\tau=t} \sqrt{\mathbf{r}'(\tau) \cdot \mathbf{r}'(\tau)} d\tau$$

τ is simply there so we do not confuse the limits of integration with the integral. For a helix, we find that:

$$\mathbf{r}' \cdot \mathbf{r}' = a^2 + c^2 = K^2$$

K is the total curvature of the helix. Then:

$$s(t) = \int_0^t \sqrt{\mathbf{r}' \cdot \mathbf{r}'} d\tau = \int_0^t K d\tau = Kt$$

Consider a parabola $y = x^2$. We can easily parametrize this with $x = \tau, y = \tau^2$:

$$\mathbf{r}(\tau) = [\tau, \tau^2]$$

We can parametrize any function like this. Then:

$$l = \int_0^t \sqrt{\mathbf{r}' \cdot \mathbf{r}'} d\tau = \int_0^t \sqrt{1 + 4\tau^2} d\tau = \frac{1}{2}t\sqrt{1 + 4t^2} - \frac{1}{4}\log(-2t + \sqrt{1 + 4t^2})$$

Points on $y = x^2$ are not necessarily spaced out equally along the curve, but points in units of $s(t)$ will be equally spaced out along $l = y$. Even though it is not always easy to find $s(t)$, it still gives

us some general important insights. We can relate $\mathbf{r}'(t)$ to $\mathbf{r}'(s)$ using the chain rule:

$$\mathbf{r}'(t) = \mathbf{r}'(s) \frac{ds(t)}{dt}$$

But what is $s'(t)$?

$$\frac{ds}{dt} = \frac{d}{dt} \int |\mathbf{r}'| dt = |\mathbf{r}'(t)| \implies \boxed{\mathbf{r}'(s) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}}$$

So we find that $\mathbf{r}'(s)$ is a unit vector, which justifies why the points as a function of s along the curve will be equally spaced out. $\mathbf{r}'(s)$ is the unit tangent vector, $\mathbf{u}(t)$.

The tangent vector can be thought of as a velocity:

$$\mathbf{r}'(t) = \frac{dx}{dt} \mathbf{i} + \frac{dy}{dt} \mathbf{j} + \frac{dz}{dt} \mathbf{k}$$

So $\mathbf{r}'(t)$ is the velocity vector along the 2D or 3D curve. The magnitude of $\mathbf{r}'(t)$ is the speed along the curve:

$$\mathbf{r}'(t) = |\text{speed}| \mathbf{u}(t)$$

So essentially, $\mathbf{r}'(s)$ removes the speed from the parametrization and only gives us the direction of the velocity.

What about curvature ? The curvature is the rate at which the angle of the tangent vector to the curve changes. Of course, this also depends on *how* we move along the curve. So the curvature of the curve is related to the second derivative of the curve.

The curvature of a 1D function is defined as:

$$\kappa(x) = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$$

For 2D or 3D:

$$\kappa(t) = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$$

Can we express this in terms of s ? κ is the rate of change of the unit tangent with s :

$$\boxed{\kappa(s) = |\mathbf{u}'(s)| = |\mathbf{r}''(s)|}$$

So the curvature is simply the second derivative of the curve as a function of arc length. For a circle:

$$\mathbf{r}(t) = [a \cos t, a \sin t] \implies \mathbf{r}'(t) = [-a \sin t, a \cos t]$$

$$s(t) = \int_0^t \sqrt{\mathbf{r}' \cdot \mathbf{r}'} d\tau = at$$

Then the curvature is $\kappa = 1/a$ (calculate on our own). The curvature has units of 1/length. This is not intuitive. So we define what is called the radius of curvature:

$$R_c = \frac{1}{\kappa}$$

It is the radius of the circle that just osculates the curve at a point, and best approximates its local curvature at that point. So small R_c means greater curvature and bigger R_c means less curvature.

9.3 Curvature and Torsion

To understand torsion, we introduce a local coordinate system which rotates and twists with the curve. This is a Frenet coordinate system. We start from the unit tangent vector $\mathbf{u}(s) = \mathbf{r}(s)/|\mathbf{r}(s)|$. There is an entire plane, normal to this vector. So we define the Frenet coordinate using the planes normal to our tangent vector. The principal normal vector is defined as:

$$\mathbf{p}(s) = \frac{\mathbf{u}'(s)}{|\mathbf{u}'(s)|} = \frac{1}{\kappa(s)} \mathbf{u}'(s)$$

Then $\mathbf{u}(s)$ and $\mathbf{p}(s)$ define the osculating plane. The binormal unit vector is given by taking the cross product of these two vectors:

$$\mathbf{b}(s) = \mathbf{u}(s) \times \mathbf{p}(s)$$

These vectors $\mathbf{u}(s), \mathbf{p}(s), \mathbf{b}(s)$ are the Frenet coordinate system.

The torsion of the curve is defined as the twisting of the osculating plane:

$$\tau(s) = -\mathbf{p}(s) \cdot \mathbf{b}'(s)$$

The negative sign is due to the definition of the direction (CW or CCW) of torsion.

9.4 Curves In Mechanics (Physical Trajectories)

Recall that $v = dx/dt = x'$ and $a = dv/dt = v' = x''$. For 3 dimensions, these definitions still hold. However, we cannot parametrize our definitions of acceleration and velocity to obtain $\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$. We know the arc length function, so we can redefine our position vector as $\mathbf{r}(s(t)) \equiv \mathbf{r}(t)$ where $s(t) = \int \sqrt{\mathbf{r}' \cdot \mathbf{r}'} dt$. So

$$\mathbf{v}(t) = \mathbf{r}'(s) \frac{ds}{dt} = \mathbf{u}(s) \frac{ds}{dt}$$

The acceleration then is

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \mathbf{u}'(s) \left(\frac{ds}{dt} \right)^2 + \mathbf{u}(s) \frac{d^2s}{dt^2}$$

So even though the velocity is always tangent, the acceleration is dependent on two components. There is the tangent acceleration which is tangent to the path and a normal acceleration to the path. Imagine a body rotating at a constant rate in a circle of radius R and angular velocity ω . The path is defined by the parametrization:

$$\mathbf{r}(t) = R \cos \omega t \mathbf{i} + R \sin \omega t \mathbf{j}$$

We can calculate the velocity as:

$$\mathbf{v}(t) = \mathbf{r}'(t) = -R\omega \sin \omega t \mathbf{i} + R\omega \cos \omega t \mathbf{j}$$

The magnitude is $R\omega$. THE acceleration is given by $\mathbf{a}(t) = -\omega^2 \mathbf{r}(t)$. This is pointing inwards. If the velocity was $\mathbf{w}(t)$ then we would have a tangential acceleration.

Imagine shooting a projectile from the north pole to some point in the equator. Due the earth spinning, there is a deflection in the path of the projectile and so it will not go straight down! This is the Coriolis force. Let us define the problem. A projectile P moves with constant speed along a meridian of the rotating earth. Consider a unit vector $\mathbf{b}(t)$ which rotates along the z axis of the earth:

$$\mathbf{b}(t) = \cos \omega t \mathbf{i} + \sin \omega t \mathbf{j}$$

The meridian plane is a plane consisting of $\mathbf{b}(t)$ and $\mathbf{k}$. This is the plane along which the projectile moves. Suppose it is moving at a constant angular rate γ along its meridian plane. The position vector is then given by:

$$\mathbf{r}(t) = R \cos \gamma t \mathbf{b}(t) + R \sin \gamma t \mathbf{k}$$

Where R is the radius of the earth. Again, γ is the angle at which the projectile travels on the surface. Then

$$\mathbf{v}'(t) = R \cos \gamma t \mathbf{b}' - \gamma R \sin \gamma t \mathbf{b} + \gamma R \cos \gamma t \mathbf{k}$$

The acceleration of our projectile is then:

$$\mathbf{a}(t) = R \cos \gamma t \mathbf{b}'' - 2R\gamma \sin \gamma t \mathbf{b}' - \gamma^2 R \cos \gamma t \mathbf{b} - \gamma^2 R \sin \gamma t \mathbf{k} = R \cos \gamma t \mathbf{b}'' - 2R\gamma \sin \gamma t \mathbf{b}' - \gamma^2 R \mathbf{r}$$

The derivatives of $\mathbf{b}$ are:

$$\mathbf{b}' = -\omega \sin \omega t \mathbf{i} + \omega \cos \omega t \mathbf{j} \quad \mathbf{b}'' = -\omega^2 \mathbf{b}(t)$$

Combining these we get:

$$\boxed{\mathbf{a}(t) = -\omega^2 R \cos \gamma t \mathbf{b}(t) - 2\gamma R \sin \gamma t \mathbf{b}' - \gamma^2 \mathbf{r}(t)}$$

The centripetal acceleration is one $-\omega^2 R \cos \gamma t \mathbf{b}(t)$ which is towards the earth's axis due to rotation of earth. And there is $-\gamma^2 \mathbf{r}(t)$ which is the centripetal force towards the earth's center. However, there is a third interaction term, $-2\gamma R \sin \gamma t \mathbf{b}'$ which is the Coriolis effect which makes it look like the projectile is deviating from its path. This is the Coriolis force.

10 Vector Fields

Consider the set of points $(x, y) \in \mathbb{R}^2$. Suppose the velocity of wind at any point in $\mathbb{R}^2$ is given by the vector $[y, x]$. Then we can express this vector function as:

$$\mathbf{F}(x, y) = [y, x] = y\mathbf{i} + x\mathbf{j}$$

We can also express $\mathbf{F}$ as a function of some curve $\mathbf{r} = [x, y]$ such that $\mathbf{F}(\mathbf{r}) = [y, x]$. A vector field, $\mathbf{F}$, is a function whose value at a point (x, y) is a vector. In general:

$$\mathbf{F}(x, y, z) = \mathbf{F}(\mathbf{r}) = F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k} = F^\mu(x, y, z)e_\mu$$

Where the F_i are scalar functions. Vectors along a parametrized surface or curve are also vector fields!

11 Gradients and Directional Derivatives

consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ where the parameters of f , x and y are themselves single variable functions $x, y : \mathbb{R} \rightarrow \mathbb{R}$ such that $x = x(t)$, $y = y(t)$. This means we can use the chain rule to calculate df/dt ! Now if we are defining the derivative, we can define a step h in the direction $\vec{u}$ as u_1h and u_2h . This yields the idea of a directional derivative:

$$D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + u_1h, y + u_2h) - f(x, y)}{h}$$

Since $\mathbf{u}$ is a unit vector, moving 1 step in time means 1 step in $\mathbf{u}$. So parametrization of our vector allows us to calculate directional derivatives. In general:

$$D_{\mathbf{u}}f = \partial_i f^i u^i = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] \cdot \frac{\mathbf{u}}{|\mathbf{u}|}$$

The gradient operator is intimately tied to directional derivative. the directional derivative. We define the gradient vector operator:

$$\nabla \equiv \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]$$

So the directional derivative is simply the projection of ∇f onto $\mathbf{u}$. So:

$$\boxed{D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}}$$

The gradient vector is a vector field. It is always orthogonal to the isolines (or isosurfaces) of the function. Gradient vectors are also independent of the coordinate system we choose! Vector

components change when we change coordinate system, but once put together, the vector does not change its magnitude or direction!

Let us explore the directionality of the gradient vector:

$$D_u f = \nabla f \cdot \mathbf{u} \implies D_u f = |\nabla f| |\mathbf{u}| \cos \theta$$

If $\cos \theta = 0$ then $D_u f$ is 0. So if f is not changing its value in the direction of $\mathbf{u}$, the gradient is perpendicular to the direction. But that is precisely the definition of the isoline or isosurface! This is why the gradient is always normal to isolines.

Now, we know that scalar functions increase the most (by the above logic) when ∇f is parallel to $\mathbf{u}$.

So by definition:

$$\frac{\nabla f}{|\nabla f|} = \mathbf{u}$$

But recall

$$D_u f = \nabla f \cdot \mathbf{u} = \frac{|\nabla f|^2}{|\nabla f|} = |\nabla f|$$

Thus the directional derivative achieves its maximum value when in the direction of the gradient, and its maximum value will always be the magnitude of the gradient vector at that point.

When dealing with 3D objects we often need to determine surface normal vectors. We can find a relationship between the surface normal vector and the gradient. Consider an isosurface $M : f(x, y, z) = \text{const}$. Let $C : \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ passing through P be a curve such that $C \in M$. The tangent to this curve is $\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}$. The tangent plane is the plane of all tangent vectors of all curves passing through P. The vector normal to the tangent plane is called the surface normal, $\mathbf{n}$. This is also normal to any curve that passes through P. A curve can be defined by $f(x(t), y(t), z(t)) = \text{const}$. Using chain rule, we find:

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = 0 \implies \boxed{\nabla f \cdot \mathbf{r}' = 0}$$

So the gradient of f is normal to the curve's tangent! Thus, the gradient is the surface normal $\mathbf{n}$.

So any vector function/field can be 'created' from some scalar function since $f(x, y, z)$ is a function

but its gradient is a vector field. If we take this the other way, if a vector fields can be the described as the gradient of a scalar function, then that scalar function is called the potential function of the field.

12 Divergence and Curl

12.1 Divergence

Given a vector field $\mathbf{v} = v_1(x, y, z)\mathbf{i} + v_2(x, y, z)\mathbf{j} + v_3(x, y, z)\mathbf{k}$, the divergence operator is defined as :

$$\text{div}(\mathbf{v}) = \nabla \cdot \mathbf{v} = \sum_{\nu} \frac{1}{h_{\nu}} \partial_{\mu} v^{\mu}$$

where $\text{div} : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_{\mu\nu} = \delta_{\mu}(h_{\nu})^2$. All formulas for gradient and divergence are similar to differentiation. If $\text{div}(\mathbf{v}) = 0$ then the flow of the field in and out is the same. If $\text{div}(\mathbf{v}) > 0$ then the strength is increasing, the vectors are moving away from the source. If $\text{div}(\mathbf{v}) < 0$ strength is decreasing and vectors are flowing towards the source. An application is in fluid dynamics. Consider the quantity mass flow rate: $m = \rho v$ where ρ is density and v is the velocity of the water. In a unit volume δy , the mass flow in is:

$$m_x = \rho v_1(x, y) \delta y$$

The mass flow out is then:

$$m_{x+\delta x} = \rho v_1(x + \delta x, y) \delta y$$

So

$$\delta m_x = \rho [v_1(x + \delta x, y) - v_1(x, y)] \delta y$$

If there is mass conservation, then: $m_{net} = 0$. So $m_x + m_y = m_{x+\delta x} + m_{y+\delta y} = 0$. Using Taylor series,

$$f(x + \delta x, y) - f(x, y) \approx \frac{\partial f}{\partial x}$$

So mass conservation becomes:

$$\rho \delta x \delta y [\nabla \cdot \mathbf{v}] = 0 \implies \nabla \cdot \mathbf{v} = 0$$

This is called the continuity equation! Our first partial differential equation, this is the principle of the conservation of mass. The actual continuity equation looks like this:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \mathbf{v})$$

This is when fluid is compressible. In the above, we assumed incompressible fluid, so density is constant.

12.2 Curl

Recall the divergence of a vector field is defined as:

$$\text{div}(\mathbf{v}) = \nabla \cdot \mathbf{v} = \partial_\mu v^\mu$$

The curl is defined as the cross product of the gradient operator with the vector field:

$$\text{curl}(\mathbf{v}) = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

It is analogous to a cross product - it yields a vector. But what does it mean?

Consider a rigid body rotation velocity field:

$$\mathbf{v} = \omega \times \mathbf{r} = -\omega y \mathbf{i} + \omega x \mathbf{j}$$

Then what is the curl ?

$$\nabla \times \mathbf{v} = 2\omega \mathbf{k}$$

So the curl is pointing out of the vector field! It is indeed perpendicular. If ω was 0, then the curl would be zero, so there's something between rotation and curl!

More generally, the curl tells us the direction of the axis of rotation of a vector field. The magnitude is the strength of the rotation or the rate of flow about that axis. Now how do we define a positive or negative rotation for $\mathbb{R}^3$? Well, we can always break this vector down to its components where a signed rotation is defined! A note for later; irrotational vector fields are those with 0 curl. This

will have implications for line integrals later.

Remember, curl is invariant under a coordinate transformation!

13 Line Integrals

Recall $\int_a^b f(x) dx$ is the area under the curve in 2D. A line integral is a generalization of this. It is the area under an n -dimensional curve.

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

C is a piecewise smooth path of integration, where $\mathbf{F}(\mathbf{r})$ is a vector function and $d\mathbf{r}$ is a differential element along $\mathbf{r}$.

Suppose we know the forces on an object exerted by air. We've learned that $W = \mathbf{F} \cdot \mathbf{r}$. Then we can write the component of $\mathbf{F}$ in the direction of $d\mathbf{r}$ as its projection. This is assuming that $\mathbf{F}$ is constant and C is a straight line. Now, suppose we have an arbitrary piecewise smooth path, C that is parametrized by $\mathbf{r}$ and the wind field is given by the vector field $\mathbf{F}(x, y)$. Then the total work is the cumulation of all infinitesimal dot products of $\mathbf{F}$ with $d\mathbf{r}$. It is essentially taking a Riemann sum along a path.

$$\delta\mathbf{r}_i = \mathbf{r}_i - \mathbf{r}_{i-1}$$

Then $\delta W = \mathbf{F}(\mathbf{r}_i) \cdot \delta\mathbf{r}_i$. And so:

$$W = \lim_{||p|| \rightarrow 0} \sum_{i=0}^N \mathbf{F}(\mathbf{r}_i) \cdot \delta\mathbf{r}_i = \int_C \mathbf{F} \cdot d\mathbf{r}$$

And this is the definition of a line integral! If we integrate over a closed curve, we put a circle on the integral. Suppose C is a closed path, then the integral is denoted by

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

We use this to define circulation and rotation. This does seem to be related deeply to the curl! In fact, we find (later) that the circulation of conservative vector fields is 0! But how do we evaluate these line integrals?

We must indeed go back to parametrization. Suppose C is parametrized by a curve $\mathbf{r}(t)$. Then a

differential element of this path is (Using the Pythagorean theorem):

$$d\mathbf{r} = dx \mathbf{i} + dy \mathbf{j}$$

Therefore, our line integral is simply:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \lim_{N \rightarrow \infty} \sum_{i=1}^N \mathbf{F}_i \cdot d\mathbf{r}_i = \int_C F_1 dx + F_2 dy + F_3 dz$$

where $\mathbf{F} = \mathbf{F}(\mathbf{r}(t))$ and $d\mathbf{r} = d\mathbf{r}/dt \cdot dt$. But how do we convert the limits of the integral? This *differential* form in that sense is not really that useful. Using the chain rule on $\mathbf{r}$ we can parametrize each differential element. Then the limits of integration will be in terms of t and each differential element becomes a time derivative and the whole integral is with respect to t .

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C F_1 dx + F_2 dy + F_3 dz = \int_{t_0}^{t_1} \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt = \int_a^b \mathbf{F} \cdot \mathbf{r}' dt$$

Where each component of $\mathbf{F}$ itself is a function of $\mathbf{r}$. This is a much more handy representation. Visually, a line integral is like the area of a curtain hanging from our curve down to the perpendicular plane.

The choice of parametrization for C or $\mathbf{r}$ does NOT matter. Line integrals change with paths. Just because a two paths start and end at the same points, does not mean their line integrals are equal! This is easily proved by contradiction. Line integrals are usually path dependent.

13.1 Path Independence

Sometimes, line integrals can be path independent. This essentially means that our integral has the same value from point A to B regardless of what path it takes. An example of this is the work done, and this goes largely to the vector field's property. Recall that if a field is conservative, then we have:

$$F^i = \partial_i f$$

Assuming f is differentiable. If we use this for the line integral formulation, we have:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} dt = \int_C df = f(B) - f(A)$$

This is the fundamental theorem of line integrals. We can find f for $\mathbf{F}$ using the exactness approach.

What does this imply for closed curves ? We can divide our curve, $C = C_1 + C_2$. Then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_A^B \mathbf{F} \cdot d\mathbf{r} + \int_B^A \mathbf{F} \cdot d\mathbf{r} = 0$$

Thus, if $\mathbf{F}$ is a conservative vector field, then there is path independence which means that the line integral over a simply closed curve is always 0. **Note:** the domain must be *simply connected*! If $\mathbf{F}$ is not differentiable over the closed interval, you cannot use this!

Some examples of path independence are why there are no magnetic monopoles or how the work done over a closed curve is 0 since no energy is lost. On the other hand, for non-conservative or dissipative vector fields, energy may be lost and so the integral will be path dependent.

13.2 Scalar Line Integrals

Recall:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F}(\mathbf{r}) \cdot \mathbf{r}'(t) dt$$

This integrand is scalar! So line integrals are in some way related to scalar integrals. The idea of a line integral is to integrate a scalar density over some curve. Then:

$$A = \sum_{i=1}^n f(r_i) \Delta \mathbf{r}_i = \sum_{i=1}^n f(r_i) \Delta s_i$$

Where Δs is just the arc length differential. As $n \rightarrow \infty$, this becomes an integral:

$$A = \int_C f ds$$

Note, with vector line integrals, direction of the path mattered, but for scalar line integrals;

$$\int_C f ds = \int_{-C} f ds$$

For a parametrized path, we can reformulate the line integral as:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$$

Suppose now we parametrize our curve by arc length, $\mathbf{r}(s)$. Then:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt = \int_a^b f(\mathbf{r}(s)) |\mathbf{r}'(s)| \, ds = \int_a^b f(\mathbf{r}(s)) \, ds$$

14 Green's Theorem

Green's theorem provides a connection between line integrals and double integrals. Stokes' connects line and surface integrals, and Gauss' (or Divergence) theorem connects surface and volume integrals. According to Green's theorem:

$$\oint_C F_1 \, dx + F_2 \, dy = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx \, dy$$

R does not need to be simply connected and the integration over C must be integrated counterclockwise. Note that here we must have that $C = \partial R$. For a 2 dimensional vector field:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} \, dA$$

Recall that the curl is the difference of the reciprocating partial derivatives in the direction of $\mathbf{k}$, thus, multiplying by $\mathbf{k}$ gives a scalar.

Intuitively, a vector field's tendency to rotate in a manner consistent with the orientation of the closed curve is the field's circulation. Green's theorem essentially calculates the net circulation about a closed curve. The curl of $\mathbf{F}$ is essentially the circulation density of the vector field at some point. Then Green's theorem is saying that to find the total circulation, we sum over every piece, dA , multiplied by its circulation density, $\nabla \times \mathbf{F}$.

Consider the area of a planar region. We take the vector field $\mathbf{F} = [-y, x]$, then the area of this region:

$$\iint_R dA = \frac{1}{2} \oint_C x \, dy - y \, dx$$

Note that orientation of C must be counterclockwise. The reason why we pick $\mathbf{F} = [-y, x]$ is so that $\nabla \times \mathbf{F} = \mathbf{k}$. It could technically be any vector field that satisfies unit circulation.

We can also connect Green's theorem to the Laplacian.

In fact, we know that if $\mathbf{F} = \nabla f \implies \nabla \times \mathbf{F} = \nabla \times \nabla f = 0$. which proves $\oint_C \nabla f \cdot d\mathbf{r} = 0$. Suppose $w : \mathbb{R}^2 \rightarrow \mathbb{R}$, then:

$$F_1 = -\frac{\partial w}{\partial y}, F_2 = \frac{\partial w}{\partial x} \implies \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \nabla^2 w \, dA = \oint_C \frac{\partial \mathbf{F}}{\partial \mathbf{n}} \, ds$$

This is important for heat conduction. If w is the temperature field, we find:

$$\iint_R \nabla^2 T \, dA = \oint_C (\nabla T \cdot \mathbf{n}) \, ds \implies \iint_R (-K \nabla^2 T) \, dA = \oint_C (\mathbf{q} \cdot \mathbf{n}) \, ds$$

And thus we have Fourier's law: $\mathbf{q} = -k \nabla T$. If there is no heat flux, then there is conservation of heat. And evidently, this leads to Laplace's equation $\nabla^2 T = 0$ which we know is the condition for a conservative vector field, $\mathbf{q}$.

In summary, integrating the circulation density of a **smooth** vector field (partial derivatives are C^1) on a region bounded by a **simple** closed curve gives the total circulation of the vector field around the boundary of the region with suitable orientation.

15 Surfaces and Surface Integrals

15.1 Surfaces

Torus parametrization:

$$\begin{bmatrix} (b + a \cos \theta) \cos \phi \\ (b + a \cos \theta) \sin \phi \\ a \sin \theta \end{bmatrix}; \theta, \phi \in [0, 2\pi]$$

Consider a manifold $\mathcal{M}$ and some curve $\mathbf{r}_c(t) \in \mathcal{M}$. We know that we can parametrize the entire manifold as $\mathbf{r}(u, v)$. Then we can describe our curve as $\mathbf{r}(u(t), v(t))$. The tangent vector is then:

$$\mathbf{r}'_C(t) = \frac{\partial \mathbf{r}}{\partial u} \frac{du}{dt} + \frac{\partial \mathbf{r}}{\partial v} \frac{dv}{dt} = \mathbf{r}_u \frac{du}{dt} + \mathbf{r}_v \frac{dv}{dt}$$

So we have the span of orthogonal vectors $\mathbf{r}_u$ and $\mathbf{r}_v$ which form our tangent plane. Then using that, we can find the normal vector to our 3 dimensional surface, as:

$$\mathbf{N} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$$

And thus, we can figure out the unit surface normal as:

$$\mathbf{n} = \frac{\mathbf{N}}{|\mathbf{N}|} = \frac{\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}}{\left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right|}$$

However, we earlier found that the unit normal of a scalar field can be defined as

$$\mathbf{n} = \frac{\nabla f}{|\nabla f|}$$

But the power of the above definition is when we are actually interested in the magnitude of normals change. Surface integrals are about calculating the analog of line integrals but for smooth (or piecewise smooth) surfaces, with continuous normal fields. Instead of the tangent vector $\mathbf{r}'(t)$, we use the unit normal, $\mathbf{n}(u, v)$. If we already have the surface function, we use the gradient definition. If we have a difficult surface, we can parametrize it and use the parametric definition to convert the surface integral into a definite double integral, similar to converting a line integral into a single integral.

Consider a parametrized surface, $\mathbf{r}(u, v)$ over some region (or domain) D such that

$$D := \{(u, v) \in \mathbb{R}^2 | u \in [a, b], v \in [c, d]\}$$

Then, we can calculate $\mathbf{r}_u$ and $\mathbf{r}_v$, and then use Riemann sums (there are smarter ways which will be discovered in MIE235) to find the surface area of our surface over D .

$$SA = \sum_{i \in [1, m]} \sum_{j \in [1, n]} |\mathbf{r}_u(u_i, v_j) \times \mathbf{r}_v(u_i, v_j)| \Delta u \Delta v$$

Thus, the surface area can be written as the following integral:

$$SA = \iint_D \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$

Another way to express this is using the function's formula: $z = f(x, y)$. Then:

$$SA = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dA$$

15.2 Surface Integrals

Consider a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and a surface $S \subset \mathbb{R}^3$. What is the flow of $\mathbf{F}$ through S is the question of interest. Suppose $\mathbf{n}$ is the surface's normal vector. Then the flux is given by

$$\phi = \iint_{S^*} \mathbf{F} \cdot \mathbf{N} dA$$

Consider a parametrization of $S = \mathbf{r}(u, v)$. Then:

$$\Delta\phi = \mathbf{F} \cdot \mathbf{n} \Delta SA = \mathbf{F} \cdot \mathbf{n} \cdot |\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v = \mathbf{F} \cdot \mathbf{N} \Delta u \Delta v \implies \phi = \iint_{S^*} \mathbf{F} \cdot \mathbf{N} dA$$

considering the direction cosines, $\mathbf{n} = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$. Then:

$$\phi = \iint_{S^*} (F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma) dA = \iint_{S^*} F_1 dy dz + F_2 dx dz + F_3 dx dy$$

Note that S^* is the projected form of S onto the (u, v) domain. The role of orientation is then shown in swapping $\mathbf{r}_u \times \mathbf{r}_v$ with $\mathbf{r}_v \times \mathbf{r}_u$ which will just change the sign of $\mathbf{N} \rightarrow -\mathbf{N}$. Thus, a clockwise orientation will just introduce a negative sign in the integral. Orientable manifolds have a continuous normal across the whole manifold. So:

$$\exists \mathbf{N} : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \text{ s.t. } \mathbf{x} \cdot \mathbf{N} = 0 \quad \forall \mathbf{x} \in S, \mathbf{N} \in C^2 \iff S \text{ is Orientable}$$

15.3 Scalar Surface Integral

Suppose $f(\mathbf{r}) = \mathbf{F} \cdot \mathbf{r}$ is the density over a surface. Then, the scalar surface integral gives the accumulative work done over some path on the surface, S .

$$\begin{aligned} \Delta W &= f(\mathbf{r}) \Delta SA = f(\mathbf{r}) |\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v = |\mathbf{N}| \Delta u \Delta v \\ \implies \delta W &= \sum_{i=1}^m \sum_{j=1}^n f(\mathbf{r}) |\mathbf{N}| \Delta u \Delta v \quad \left[\therefore W = \iint_D f(\mathbf{r}) |\mathbf{N}| du dv \right] \end{aligned}$$

Where D is the projection of S onto the (u, v) space. For unit density $f(\mathbf{r}) = 1$, the integral simplifies to the formula for surface area:

$$SA = \iint_D |\mathbf{N}| du dv$$

We can integrate any scalar quantity we want. The mass moment of inertia of a lamina S of mass M about some axis L with density μ and distance function to L given by D , then:

$$I = \iint_S \mu(\mathbf{r})[D(\mathbf{r})]^2 dA = \iint_S \mu D^2 dA = \iint_D \mu D^2 |\mathbf{N}| du dv$$

is the mass moment of inertia.

16 The Divergence (Gauss') Theorem

16.1 The Divergence Theorem

The divergence theorem connects surface integrals and volume integrals. Suppose we have a closed volume element V and its closed surface, S such that $S = \partial V$. Then for a vector field $\mathbf{F} : V \rightarrow \mathbb{R}^3$. Then we must have that:

$$\oiint_{\partial V} \mathbf{F} \cdot \mathbf{n} dA = \oiint_S \mathbf{F} \cdot \mathbf{n} dA = \iiint_V \nabla \cdot \mathbf{F} dV$$

If $\mathbf{F}$ has a divergence such that $f(\mathbf{r}) = \nabla \cdot \mathbf{F}$, then we have that:

$$\iiint_V f(\mathbf{r}) dV = \oiint_{\partial V} \mathbf{F} \cdot \mathbf{n} dA$$

Proof follows from generalized Stokes' theorem. Conceptually, we are breaking up V into infinitely many pieces, dV and calculating the flux over each one, and then summing over them. Consider the divergence theorem in 2 dimensions of space:

$$\oiint_R \nabla \cdot \mathbf{F} dA = \oint_{\partial R} \mathbf{F} \cdot \mathbf{n} ds = \oint_C \mathbf{F} \cdot \mathbf{n} ds$$

such that $\mathbf{F}$ is a vector field as a function of $\mathbf{r}(s)$ where s is the arc length. If we recall the definitions, we know that $\mathbf{r}'(s) = dx/ds\mathbf{i} + dy/ds\mathbf{j}$ and $\mathbf{n} = dy/ds\mathbf{i} - dx/ds\mathbf{j}$. Then, we have that:

$$\oint_R \frac{dF_1}{dx} + \frac{dF_2}{dy} dA = \oint_{\partial R} F_1 \frac{dy}{ds} ds - F_2 \frac{dx}{ds} ds \implies \oint_R \frac{dN}{dx} - \frac{dM}{dy} dA = \oint_{\partial R} N dy - M dx$$

So the Divergence theorem is the same thing as Green's theorem, but in 3 dimensions instead of 2. We will also show that Stokes' theorem is also just a generalized version of Green's and Gauss' theorems. Sometimes, volumes are harder to mesh compared to surfaces, so using surface integrals could be easier for FEA/CFD.

16.2 Divergence Theorem, Heat Conduction PDE and Fourier Series

Recall from Green's theorem, we derived a PDE for steady state 2 dimensional heat conduction:

$$\iint_R (-k\nabla^2 T) dA = \oint_C (\mathbf{q} \cdot \mathbf{n}) ds \implies \mathbf{q} = -k\nabla^2 T$$

where k is the thermal conductivity. Now, consider instead of a 2 dimensional region, a 3 dimensional surface! Then:

$$\iint_S \mathbf{q} \cdot \mathbf{n} dA = \iiint_V \nabla \cdot (-k\nabla T) dV = - \iiint_V k\nabla^2 T dV$$

The total heat stored in a volume, can be calculated as:

$$H = \iiint_T \rho C T dV [=] J$$

where ρ is the density and C is the specific heat of the material. Conservation of energy says that the decrease of H must be equal to the amount of heat leaving the surface:

$$\implies -\frac{\partial H}{\partial t} = - \iiint_T \rho C \frac{\partial T}{\partial t} dV$$

We also know that:

$$\iint_S \mathbf{q} \cdot \mathbf{n} dA = - \iiint_T k\nabla^2 T dV$$

So heat flux must equal change in H implies that the right hands must be equal:

$$\iiint_T \rho C \frac{\partial T}{\partial t} dV = \iiint_T k \nabla^2 T dV$$

And this gives us the heat equation PDE:

$$\frac{\partial T}{\partial t} = \alpha^2 \nabla^2 T \quad \text{s.t.} \quad \alpha^2 = k/\rho C$$

α is in some sense similar to the Reynolds number for fluids but it is applied to heat conduction.

17 Stokes' Theorem

Stokes' theorem, similar to Green's theorem, provides a connection between surface integrals and line integrals. Let $\mathbf{F}$ be a vector field over some manifold, S . Recall green's theorem For a region $R \subset \mathbb{R}^2$:

$$\iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA = \oint_{\partial R} \mathbf{F} \cdot d\mathbf{r}$$

This is indeed similar to Stokes' theorem.

It states for a manifold S :

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} dA = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$$

such that S is piecewise smooth **open** surface.

Note that in divergence theorem, it needed to be a closed surface. With Stokes' theorem, we can use it for open surfaces! It transforms surface integrals over manifolds into the line integral over their contour. Suppose $\mathbf{F}$ is a conservative vector field. Then, we must have that $\mathbf{F} = \nabla f$. Then Stokes' theorem says:

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} dA = \iint_S \nabla \times \nabla f \cdot \mathbf{n} dA = \oint_{\partial S} \nabla f \cdot d\mathbf{r}$$

But, we have that $\nabla \times \nabla f = 0$. Thus,

$$\mathbf{F} \text{ is a conservative field} \iff \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = 0$$

Intuitively, this is when the rotations densities of each sub region in S cancel out, meaning the entire flux is 0.

Similarly, if we consider a 2 dimensional vector field:

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}$$

then we have that:

$$\iint_S \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k} \cdot \mathbf{n} dA = \iint_S \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} dA = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$$

And this is just Green's theorem. So Green's theorem is simply Stokes' theorem but in 2 dimensions.

Really the entirety of calculus is in the generalized Stokes' theorem:

$$\int_{\partial S} \omega = \int_S d\omega$$